PROJECT SUTRA — HARIKA'S PITCH SCRIPT

SPEAKER: HARIKA • OPERATIONS LEAD • SLIDES 01 & 02

TOTAL TIME: ~65 SECONDS
STAGE: CENTER STAGE

💡 **DELIVERY TIPS FOR HARIKA:** Speak clearly, naturally, and with urgent conviction. Maintain confident eye contact with the jury. You set the entire high-stakes mood of the room before handing over to the engineering leads.

SLIDE 01 Opening Hook & Project SUTRA Introduction

00:00 – 00:30 (30s)

[Stand center stage • Take a 1-second pause • Deliver opening hook with serious, grounded tone]

🔥 **HOOK: "When over 300 lives were buried under mud in the July 2024 Wayanad landslides, and flash floods cut off entire mountain valleys across Kedarnath and Nepal, rescue teams had only 72 hours to save lives. But the moment standard drones entered thick forests and deep ravines, they lost signal and crashed."**

"Respected jury, we are **Team OFFGRID**. I am Harika, and with me are Vedanth and Nikhil. We built **Project SUTRA** — an autonomous drone swarm engineered to find survivors even when **GPS fails and normal communications collapse**."

[Point smoothly toward the 3 cards on Slide 1]

"SUTRA is powered by 3 core breakthroughs:"

- **PILLAR 1** **Self-Flying Swarms:** Drones dodge trees and coordinate together at 50 times a second without ground towers. `[50Hz GNC]`
- **PILLAR 2** **Smart AI Video:** Live video stays connected through deep mountain valleys without blacking out. `[-5dB JSCC]`
- **PILLAR 3** **Thermal & 3D AI Vision:** Spots survivors through smoke and finds their exact coordinates on steep slopes. `[3D DEM • 3.59cm]`

SLIDE 02 The Problem: 4 Real Disaster Breakdowns

00:30 – 01:05 (35s)

[Click to Slide 2 • Transition tone to urgent, factual investigation]

⚠️ **THE VOD: "When our team audited actual NDRF and Army rescue reports from recent disaster zones, we found 4 major breakdowns:"**

- **BREAKDOWN 1** **Trees Block GPS:** In Wayanad, thick tree cover blocked satellite signals, causing **70% of drones to crash into branches**.
- **BREAKDOWN 2** **Mountains Block Radio:** In mountain gorges, steep ridges blocked radio signals, dropping video feeds to total black.
- **BREAKDOWN 3** **Slope Errors:** On steep 45-degree hillsides, standard drone cameras assumed flat ground, **missing survivor locations by 15 to 30 meters**.
- **BREAKDOWN 4** **Pilot Overload:** Deploying 5 separate drones required **15 to 25 crew members** and 2 hours of setup, costing ₹12.5 Lakhs per flight.

🤝 **POWER HANDOFF TO VEDANTH:**
"Vedanth will now explain how our AI vision and tactical ground control solve these bottlenecks."